

A • SOURCES OF LIGHT — Natural & Artificial

Light	Form of energy that enables us to see objects — stimulates the retina of our eye
Luminous Object	Object that produces/emits its own light — e.g., Sun, stars, firefly, electric bulb, candle
Non-luminous Object	Object that does NOT produce its own light — reflects light from another source — e.g., Moon, planets
Moon	NOT luminous — reflects sunlight toward Earth; only the lit half visible from Earth

Natural Sources of Light

Sun	Primary and major source of natural light — essential for photosynthesis
Stars	Produce light just like the Sun — appear dim because they are far away
Firefly	Produces light by bioluminescence — chemical reaction inside the organism
Bioluminescence	Ability of living organisms to produce light — fireflies, jellyfish, glow worms, deep sea creatures

Artificial Sources of Light

Type of Artificial Source	Description	Examples
Incandescent Sources	Objects heated to very high temperature begin to emit light — heat → light	Candle, incandescent lamp (electric bulb), glowing hot iron rod
Gas Discharge Sources	Electricity passed through gas at very low pressure produces light — electrical discharge → light	Neon lamp (Ne gas), Sodium lamp — used in street lights
Fluorescent Sources	UV light from mercury vapour excites phosphor coating on bulb → visible light	Tube light (CFL) — gas discharge + fluorescence

B · PROPERTIES OF LIGHT — Rectilinear Propagation & Speed

Rectilinear Propagation	Light travels in a STRAIGHT LINE — cannot bend on its own path
Speed of Light	3×10^8 m/s in air or vacuum — nothing can travel faster than light
Light year	Distance light travels in one year — 9.46×10^{15} m — used for astronomical distances

- Evidence of rectilinear propagation: light through holes in matchboxes visible only when all holes align in a straight line
- Pinhole camera works because light travels in straight lines — image formed is inverted and real
- Al-Hasan al-Haytham (Arab scientist) — first to experiment with light; proved light enters eye from external source, not emitted by eye
- Solography: pinhole camera used to photograph Sun's movement over long periods

Parallel beam	Rays parallel to each other — e.g., sunlight, car headlights
Convergent beam	Rays meeting at a point — e.g., rays focused by a lens
Divergent beam	Rays spreading outward from a point — e.g., candle flame, torch

C · INTERACTION OF LIGHT WITH MATTER — Transparent, Translucent, Opaque

Type	Definition	Examples (Class 7 Textbook)
Transparent Material	Allows light to pass through COMPLETELY — objects clearly visible through it	Clear glass, eyeglasses, clear drinking glass, clear water, bus face glass, clear plastic ruler, clear water in jar
Translucent Material	Allows light to pass through PARTIALLY — objects not clearly visible through it	Rough window glass, cellophane paper, tissue paper, oiled paper, frosted glass, coconut oil, diluted milk
Opaque Material	Does NOT allow light to pass through — objects not visible through it at all	Wall, thick cardboard, stone, metal sheet, wood, aluminium foil, milk, thick coloured plastic

■ Exam Trap — Transparent vs Translucent

- Transparent: can see CLEAR image through it (eyeglasses, clear window)
- Translucent: light passes but image is BLURRED (frosted glass, tissue paper)
- Opaque: NO light passes through — forms SHADOW (wood, metal, thick cardboard)
- Same material can behave differently: thick glass = opaque; thin clear glass = transparent

D · SHADOWS — Umbra, Penumbra & Eclipses

Shadow	Dark region formed when an opaque object blocks light — shadow is on the OPPOSITE side of the light source
Umbra	Completely dark inner region of shadow — NO light reaches here — formed when point source used
Penumbra	Partially lit outer region around umbra — some light reaches here — formed with broad/extended source

Properties of Shadows

- Only OPAQUE objects form shadows (transparent objects do not form shadows)
- Shadow is always on the OPPOSITE side of the light source from the object
- The colour of the light does NOT affect the colour of the shadow — shadow is always dark
- Light source + opaque object + shadow — all in the SAME straight line
- Size of shadow depends on: (1) distance between light source and object (2) distance between object and screen
- Smaller the light source → larger the umbra (point source → no penumbra, only umbra)
- Larger the light source → smaller the umbra + larger penumbra

Solar & Lunar Eclipses

Eclipse	When it Occurs	Shadow Type	Who Experiences
Solar Eclipse	Moon comes BETWEEN the Sun and Earth — Moon's shadow falls on Earth	Umbra = Total solar eclipse (complete darkness) Penumbra = Partial solar eclipse	People in umbra region (A) = total eclipse People in penumbra (B,C) = partial eclipse
Lunar Eclipse	Earth comes BETWEEN the Sun and Moon — Earth's shadow falls on the Moon	Earth blocks sunlight from reaching the Moon	Visible from the night side of Earth — Moon appears red/dark during lunar eclipse

E · REFLECTION OF LIGHT — Laws, Types & Terms

Reflection	Bouncing back of light rays when they fall on a smooth, shiny, polished surface
Incident Ray	Ray of light falling on the reflecting surface (PO in diagrams)

Reflected Ray	Ray of light bouncing back from the reflecting surface after reflection (OQ)
Normal	Imaginary perpendicular line drawn at the point of incidence to the reflecting surface (ON)
Angle of Incidence (i)	Angle between incident ray and normal at point of incidence
Angle of Reflection (r)	Angle between reflected ray and normal at point of incidence
Point of Incidence	The exact point on the reflecting surface where the incident ray strikes (O)

Laws of Reflection

- LAW 1 — The angle of incidence (i) is ALWAYS EQUAL to the angle of reflection (r): $i = r$
- LAW 2 — The incident ray, the reflected ray, and the normal at the point of incidence all lie in the SAME PLANE
- Laws of reflection hold for ALL surfaces — smooth or rough, flat or curved
- Silver metal is the BEST reflector of light — thin silver layer deposited on glass makes a mirror

Types of Reflection

Type of Reflection	Surface	What Happens	Example	Image Formed?
Regular (Specular) Reflection	Smooth, polished, flat	Parallel rays remain parallel after reflection — each ray obeys law of reflection equally	Plane mirror, still water surface	YES — clear, sharp image
Irregular (Diffuse) Reflection	Rough, uneven	Parallel rays scatter in different directions after reflection — different normals at each point	Wall, paper, road surface	NO — no clear image; but allows us to SEE the object

■ Diffuse reflection from walls illuminates the entire room — if walls were smooth mirrors, only parts directly lit by window would be visible.

F · PLANE MIRROR — Image Properties

Properties of Image formed in a Plane Mirror — Memorise all 5

Upright (Erect)	Image is right-side up — not inverted (unlike pinhole camera which gives inverted image)
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Virtual	Image CANNOT be formed on a screen — appears to be behind the mirror
Same size as object	Image size = Object size — magnification = 1 (neither enlarged nor diminished)
Same distance behind mirror	Image distance from mirror = Object distance from mirror: $a = b$
Laterally inverted	Left and right are swapped — left hand appears as right hand in mirror; "AMBULANCE" written reversed on vehicles so drivers see it correctly in rear-view mirror

	Pinhole Camera	Plane Mirror
Image type	Real (can be formed on screen)	Virtual (cannot be formed on screen)
Orientation	Inverted (upside down)	Erect (upright)
Size	May differ from object size	Same size as object
How it works	Light passes through tiny hole	Light reflects off mirror surface

■ Did You Know?

- ◆ "AMBULANCE" written backwards on vehicle front — drivers see it correctly in their rear-view mirror due to LATERAL INVERSION.
- ◆ Multiple images in two mirrors: If angle between mirrors = θ , number of images = $(360^\circ/\theta) - 1$. For 90° : 3 images. For 60° : 5 images. For 45° : 7 images.
- ◆ When two mirrors are PARALLEL to each other, the number of images formed is INFINITE.

G · MULTIPLE REFLECTION — Kaleidoscope & Periscope

Images in 2 mirrors

$$n = (360^\circ / \theta) - 1 \quad (n = \text{number of images; } \theta = \text{angle between mirrors})$$

Device	Principle	Structure	Uses
Kaleidoscope	Multiple reflection of light between mirrors inclined to each other — creates repeated patterns	2 or 3 mirrors inclined at 60° angle, forming equilateral triangle; coloured objects (beads, bangles) inside	Toy for children; design patterns; used by artists for symmetrical design inspiration

Device	Principle	Structure	Uses
Periscope	Law of reflection — mirrors at each end of tube reflect light so observer can see over/around barriers	Long tube with 2 plane mirrors at 45° at each end; light enters top, reflects down, reflects again to eye	Submarines to see above water surface; military (bunker shooting); photographing restricted areas; doctors use fibre-optic endoscopes (similar principle)

H · SPHERICAL MIRRORS — Concave & Convex

Key Terms — Spherical Mirrors

Pole (P)	Geometric centre of the spherical mirror — where principal axis meets the mirror surface
Centre of Curvature (C)	Centre of the sphere from which the mirror is made — lies in front of concave, behind convex
Radius of Curvature (R)	Distance between the centre of sphere and the pole (vertex) of the mirror
Principal Axis	Imaginary line joining the Pole (P) and Centre of Curvature (C) of the mirror
Focus / Focal Point (F)	Point where reflected rays converge (concave) or appear to diverge from (convex) on principal axis
Focal Length (f)	Distance between the Pole (P) and the Focus (F): $f = R/2$

Focal Length Formula $f = R / 2 \rightarrow R = 2f$ (Focal length = half of radius of curvature)

Concave vs Convex Mirror — Comparison

Feature	Concave Mirror	Convex Mirror
Reflecting surface	Curved INWARD (like inside of bowl)	Curved OUTWARD (like outside of bowl)
Image — general	Real and inverted (usually); virtual and erect only when object between F and P	Always VIRTUAL, ERECT and DIMINISHED
Field of view	Narrow field of view	WIDER field of view — sees larger area
Image size	Magnified (when object close) or diminished (when object far)	Always SMALLER than object

Feature	Concave Mirror	Convex Mirror
Common uses	Make-up mirror; shaving mirror; torch/headlight; solar cooker; head mirror for ENT doctors; reflecting telescope	Rear-view mirror in vehicles; hallway safety mirrors; shop security mirrors; road curve mirrors

Image Formation by Concave Mirror — Position-wise

Object Position	Image Position	Image Size	Nature
At Infinity	At F (focus)	Highly diminished (point-sized)	Real, Inverted
Beyond C	Between C and F	Diminished	Real, Inverted
At C	At C	Same size as object	Real, Inverted
Between C and F	Beyond C	Magnified (enlarged)	Real, Inverted
At F (focus)	At Infinity	Highly magnified	Real, Inverted
Between F and P (Pole)	Behind mirror	Magnified	Virtual, Erect

■ **Concave mirror ALWAYS forms real and inverted image EXCEPT when object is between F and P — then image is virtual and erect.**

Image Formation by Convex Mirror

- Object at Infinity → Image at F (behind mirror) → Highly diminished, point-sized → Virtual and Erect
- Object between Infinity and Pole → Image between P and F (behind mirror) → Diminished → Virtual and Erect
- Convex mirror image is ALWAYS virtual, erect and diminished — regardless of object position

Applications of Curved Mirrors

Concave Mirror Uses:

- Make-up / shaving mirror — magnifies face
- Torch, search lights, headlights — focuses light to travel far
- Solar cooker — collects and focuses sunlight
- Head mirror (ENT doctors) — provides shadow-free illumination
- Reflecting telescope — collects light from stars

Convex Mirror Uses:

- Rear-view mirror in vehicles — wider field of view; objects appear closer (safety warning)
- Hallway mirrors in hospitals, hotels — safety at sharp turns
- Road curve mirrors — see traffic around bends
- Shop security mirrors — wider area surveillance
- Parabolic mirrors — solar water heaters, radio telescopes, parabolic microphones

I · REFRACTION OF LIGHT — Snell's Law & Refractive Index

Refraction	Bending of light at the normal when it passes from one transparent medium to another — due to change in speed
Rarer Medium	Medium in which light travels FASTER — e.g., air (lower optical density)
Denser Medium	Medium in which light travels SLOWER — e.g., water, glass (higher optical density)
Rarer → Denser	Light BENDS TOWARDS the normal (angle of refraction < angle of incidence)
Denser → Rarer	Light BENDS AWAY from the normal (angle of refraction > angle of incidence)

Refractive Index (μ)	$\mu = \text{Speed of light in air (c)} / \text{Speed of light in medium (v)}$ (Absolute refractive index — always > 1; no unit)
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Snell's Law	$\sin i / \sin r = \mu \text{ (constant)}$ (i = angle of incidence; r = angle of refraction)
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Snell's Laws of Refraction

- LAW 1 — The incident ray, the refracted ray and the normal at the point of incidence all lie in the SAME PLANE
- LAW 2 — The ratio of sine of angle of incidence to sine of angle of refraction = refractive index (constant) for that medium
- Refractive index has NO UNIT — it is a ratio of two speeds (same quantity)
- Refractive index of any transparent medium is ALWAYS GREATER THAN 1 (light speed in medium < light speed in air)

Refractive Index Values

Substance	Refractive Index (μ)	Relative Speed of Light
Air	1.0 (reference)	Fastest — reference medium
Water	1.33	Slower than air
Ether	1.36	Slower than air
Kerosene	1.41	Slower than air
Ordinary Glass	1.5	Much slower than air
Quartz	1.56	Much slower than air

Substance	Refractive Index (μ)	Relative Speed of Light
Diamond	2.41	Slowest — highest refractive index — that's why it sparkles!

- Why pencil in water appears bent: light travels from water (denser) to air (rarer) → bends away from normal → pencil looks bent
- Why pool appears shallower than actual depth: refraction at water-air boundary
- Why a coin invisible in empty bowl becomes visible when water added: refraction bends light toward your eye

Higher refractive index	More the bending of light; slower light travels in that medium
Diamond ($\mu=2.41$)	Light bends dramatically inside diamond → total internal reflection → sparkle effect

J · DISPERSION OF LIGHT — Spectrum, VIBGYOR & Rainbow

Dispersion	Splitting of white light into its seven constituent colours when passed through a transparent medium (prism)
Spectrum	Band of seven colours produced by dispersion: VIBGYOR
Prism	Transparent object with at least two flat surfaces forming an acute angle — refracts and disperses light
VIBGYOR	Violet-Indigo-Blue-Green-Yellow-Orange-Red — order of colours in white light spectrum
White light	Combination of ALL seven colours of visible spectrum — NOT a single colour
Black	Absence of ALL wavelengths of visible light — opposite of white

VIBGYOR — Wavelength & Deviation

Colour	Wavelength	Deviation in Prism	Speed in Medium
Violet (V)	SHORTEST (~400 nm)	MOST deviated (bends most)	Slowest in medium
Indigo (I)	Short	Highly deviated	Very slow
Blue (B)	Short	Highly deviated	Slow
Green (G)	Medium	Moderate deviation	Moderate
Yellow (Y)	Medium-long	Moderate deviation	Moderate-fast

Colour	Wavelength	Deviation in Prism	Speed in Medium
Orange (O)	Long	Less deviated	Fast
Red (R)	LONGEST (~700 nm)	LEAST deviated (bends least)	Fastest in medium

- Wavelength range of visible light: 400 nm (violet) to 700 nm (red) — $1 \text{ nm} = 10^{-9} \text{ metre}$
- Refraction is INVERSELY proportional to wavelength — shorter wavelength bends MORE
- Red light scattered LEAST by air → travels farthest through fog → used for danger signals in vehicles
- Why rainbow is seen opposite to Sun: white sunlight passing through water droplets undergoes dispersion → VIBGYOR formed
- Newton Disc: seven VIBGYOR colour sectors on rotating cardboard disc → appears white when spun fast (colour synthesis)
- Synthesis: mixing all VIBGYOR colours → white light; absence of all → black
- Second prism (inverted) after first prism recombines the seven colours back to white light
- Wavelength 400-700nm is VISIBLE LIGHT range; below 400nm = UV (invisible); above 700nm = IR (invisible)

K · OPTICAL FIBRE — Total Internal Reflection

Optical Fibre	Device based on TOTAL INTERNAL REFLECTION — transmits light signals with negligible energy loss
Total Internal Reflection	When light in denser medium hits boundary at angle > critical angle, it is COMPLETELY reflected back — does not pass through
Structure	Cable with thin flexible glass core; light travels through repeated internal reflections from one end to other
Data transmission	Information sent as pulses of light through bundles of optical fibres

- Optical fibres carry MORE data than traditional copper cable telephone lines
- Used in: cable TV, high-speed broadband internet, fibre optic communications
- Medical use: fibre-optic endoscopes used by doctors to see inside body organs (stomach, intestine)
- Periscopes sometimes use optical fibre instead of mirrors for higher resolution images
- Fibre can be twisted and bent — light still travels to the other end with negligible loss

L · Solved Numerical Problems

Problem 1	The radius of curvature of a spherical mirror is 20 cm. Find its focal length.
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Given	$R = 20 \text{ cm}$
Solution	$f = R / 2 = 20 / 2 = 10 \text{ cm}$
Answer	Focal length = 10 cm

Problem 2	Focal length of a spherical mirror is 7 cm. Find its radius of curvature.
Given	$f = 7 \text{ cm}$
Solution	$R = 2 \times f = 2 \times 7 = 14 \text{ cm}$
Answer	Radius of curvature = 14 cm

Problem 3	A light ray strikes a plane surface at 43° with the plane surface. Find: (i) angle of incidence (ii) angle of reflection (iii) angle between incident and reflected ray.
Given	Angle with surface = 43° ; Normal is perpendicular to surface
Solution	(i) Angle of incidence $i = 90^\circ - 43^\circ = 47^\circ$ (ii) By law of reflection: $r = i = 47^\circ$ (iii) $i + r = 47^\circ + 47^\circ = 94^\circ$
Answer	$i = 47^\circ$; $r = 47^\circ$; angle between rays = 94°

Problem 4	Two plane mirrors are inclined at 90° to each other. Find the number of images formed.
Given	Angle $\theta = 90^\circ$
Solution	$n = (360^\circ / \theta) - 1 = (360^\circ / 90^\circ) - 1 = 4 - 1 = 3$
Answer	Number of images = 3

Problem 5	Speed of light in air = $3 \times 10^8 \text{ m/s}$; speed in medium = $2 \times 10^8 \text{ m/s}$. Find refractive index.
Given	$c = 3 \times 10^8 \text{ m/s}$; $v = 2 \times 10^8 \text{ m/s}$
Solution	$\mu = c / v = (3 \times 10^8) / (2 \times 10^8) = 1.5$
Answer	Refractive index $\mu = 1.5$ (same as ordinary glass)

Problem 6	Refractive index of water = $4/3$; refractive index of glass = $3/2$. Find refractive index of glass w.r.t. water.
Given	$\mu(\text{water}) = 4/3$; $\mu(\text{glass}) = 3/2$
Solution	$w \mu g = \mu(\text{glass}) / \mu(\text{water}) = (3/2) / (4/3) = (3/2) \times (3/4) = 9/8 = 1.125$
Answer	Refractive index of glass w.r.t. water = $9/8 = 1.125$

M · Key Facts — Rapid Revision (Numbers, Years & Names only)

Formulas, Values & Key Numbers

Quantity / Concept	Formula / Value	Unit / Note
Speed of light	3×10^8 m/s in air/vacuum (3 lakh km/s)	Fastest possible speed
Light year	9.46×10^{15} m — distance light travels in 1 year	Astronomical unit
Visible light wavelength	400 nm (violet) to 700 nm (red)	$1 \text{ nm} = 10^{-9} \text{ m}$
Focal length formula	$f = R/2 \rightarrow R = 2f$	Spherical mirrors
Refractive index	$\mu = c/v = \text{speed in air} / \text{speed in medium}$	No unit; always > 1
Snell's Law	$\sin i / \sin r = \mu$ (constant)	Refraction law
μ of Air	1.0 (reference)	Rarer medium
μ of Water	1.33	Common in questions
μ of Glass (ordinary)	1.5	Most common in problems
μ of Diamond	2.41 (highest common substance)	Sparkle due to TIR
Number of images (2 mirrors)	$n = (360^\circ/\theta) - 1$	θ = angle between mirrors
90° mirrors	$(360/90) - 1 = 3$ images	Common exam question
60° mirrors	$(360/60) - 1 = 5$ images	Kaleidoscope typical angle
Parallel mirrors ($\theta=0^\circ$)	Infinite images	e.g., two walls of a corridor
Concave mirror — at C	Image at C; same size; real and inverted	Key position
Concave mirror — at F	Image at infinity; highly magnified; real and inverted	Key position
Concave — between F and P	Image virtual, erect, magnified (behind mirror)	Only virtual case
Convex mirror — always	Virtual, erect, diminished — regardless of object position	Key fact

Scientists, Discoveries & Years

Scientist / Event	Contribution / Fact	Year / Class
Al-Hasan al-Haytham	First to experiment with light; proved rectilinear propagation; proved light enters eye from outside (not emitted by eye)	10th–11th century
Isaac Newton	Newton Disc experiment — showed VIBGYOR colours recombine to give white light; studied dispersion with prism	1666
Diocles (mathematician)	Wrote "On Burning Mirrors" — first mention of parabolic mirrors	Greco-Roman times
Ibn Sahl (physicist)	Studied parabolic mirrors in 10th century	10th century
Heinrich Hertz	Constructed first parabolic mirrors as reflector antennae	1888
SI Unit of angle (Radian)	Plane angle = radian Solid angle = steradian	1995 reclassified

N · Glossary — Definitions

Light — Form of energy that enables us to see — stimulates the retina of the eye	Luminous Object — Object that produces its own light — e.g., Sun, stars, electric bulb, firefly
Non-luminous Object — Object that does not produce its own light — reflects light — e.g., Moon, planets	Bioluminescence — Production of light by living organisms through chemical reactions — e.g., firefly, jellyfish
Incandescent Source — Light produced by heating object to high temperature — e.g., candle, bulb	Rectilinear Propagation — Property of light to travel in a straight line
Pinhole Camera — Simple device demonstrating rectilinear propagation; forms real, inverted image through tiny hole	Transparent Material — Allows light to pass through completely — clear image visible through it
Translucent Material — Allows light to pass partially — objects visible but not clearly through it	Opaque Material — Does not allow light to pass through — forms shadows
Shadow — Dark region formed when opaque object blocks light on the opposite side of light source	Umbra — Completely dark region of shadow — no light reaches — formed with point source
Penumbra — Partially lit outer region of shadow — some light reaches — formed with extended source	Reflection — Bouncing back of light rays when they fall on a smooth, polished surface
Incident Ray — Light ray falling on the reflecting surface	Reflected Ray — Light ray bouncing back after reflection
Normal — Imaginary perpendicular line at the point of incidence on the reflecting surface	Angle of Incidence (i) — Angle between incident ray and normal at point of incidence

Angle of Reflection (r) — Angle between reflected ray and normal — always equals angle of incidence	Regular (Specular) Reflection — Reflection from smooth surface — reflected rays are parallel — forms clear image
Diffuse (Irregular) Reflection — Reflection from rough surface — reflected rays scatter — no clear image formed	Plane Mirror — Mirror with flat, polished reflective surface
Lateral Inversion — Left-right reversal in plane mirror image	Virtual Image — Image that cannot be formed on a screen — appears behind the mirror
Real Image — Image that can be formed on a screen — formed by actual convergence of light rays	Spherical Mirror — Mirror that is part of a hollow sphere — either concave or convex
Concave Mirror — Spherical mirror reflecting at inner (concave) curved surface — converging mirror	Convex Mirror — Spherical mirror reflecting at outer (convex) curved surface — diverging mirror
Focal Length (f) — Distance from pole (P) to focus (F) of a spherical mirror: $f = R/2$	Radius of Curvature (R) — Distance from centre of curvature to pole of mirror: $R = 2f$
Kaleidoscope — Toy based on multiple reflection — 2 or 3 mirrors inclined at 60° create symmetrical patterns	Periscope — Instrument using 2 mirrors at 45° to see over or around obstacles — used in submarines
Refraction — Bending of light as it passes from one transparent medium to another due to change in speed	Refractive Index (μ) — Ratio of speed of light in air to speed in medium — $\mu = c/v$; no unit; always > 1
Snell's Law — $\sin i / \sin r = \mu$ (constant) for a given medium — law governing refraction	Dispersion — Splitting of white light into VIBGYOR when passed through a transparent medium like prism
VIBGYOR — Violet-Indigo-Blue-Green-Yellow-Orange-Red — order of colours in white light spectrum	Spectrum — Band of colours produced when white light is dispersed — wavelength 400nm to 700nm
Total Internal Reflection — Complete reflection of light inside denser medium when angle $>$ critical angle — basis of optical fibre	Optical Fibre — Thin glass fibre transmitting light signals by total internal reflection — used in communication and medicine
Parabolic Mirror — Mirror with parabolic shape — focuses all parallel rays to a single focal point — used in solar cookers, telescopes	